A Robot Motion Planning Strategy For Threatening Environments

1. Introduction

The motion planning process of agents has been a heated topic in robotic fields. Many have put up algorithms which aimed at making decisions to automatically explore dynamic environments (Li et al., 2020) or achieve specific goals in a competitive game (Mnih et al., 2015). Many of them adopt deep neural networks to observe high-level environment and act as action-value function.

Inspired by the wide usage of DQN and game theory in multi-agent decision process, I hereby propose a motion planning strategy that can help agents survive while protect themselves in a dynamic environment with threatening intelligent agents, which is different from the Atari 2600 games since the "AI" in these games will not evolve by interacting with players.

2. Problem Statement

In the research, the agents are required to play a game. In the game, there will be 4 agents in the environment, and be equally divided into two hostile team, we can distinguish them by color (as figure 1). They will start at fixed start zones at the beginning of the game and attack the hostile ones in order to reduce their HP (Health Points) in the game. Agents can use barriers as shelter to protect themselves from the attack. The game will end either because of time limit or all agents in a team have 0 HP.

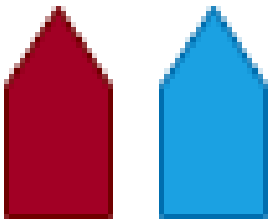

Figure 1. The icon of agents

The game has several rules as follows:

1. Agents know the coordinate and rotation angle of all 4 agents.
2. All agents are identical.
3. Agents have a weight of 20kg, a max acceleration of 2m/s^2, and a max speed of 5m/s.
4. Agent can shoot a bullet once in 1 unit time, with a max firing range of 5m, its velocity is 25m/s.
5. Agent has 400 HP in each game, they can attack other agent when it locates less than ±20° in front of the agent. the injure caused by each hit is determined by the following equation, here x denotes the distance of the agent and the target. The hit is made as soon as the decision of "shoot" aims at an agent is made. What's more, firing over barriers to hit the agents behind is not possible.

$$H(x) = 10(\frac{5}{x} - 1)$$

6. Agents will lose 10 points each time they hit the wall.
7. Agents may also harm their ally if it blocks the trajectory of bullets.

I. The statements about the environment are as follows.

   1. The environment is a 9m x 5m place with static landscape. The top-left zone ① is the startup zone for red team and the bottom-right zone ② is the blue team's.
   2. There are 3 kinds of barriers as the labels on figure 2.
      Type ③: 0.25x1m,
      type ④: 1.25x0.25m

type ⑤: 1.5x0.25m
type ⑥: 0.5x0.5m

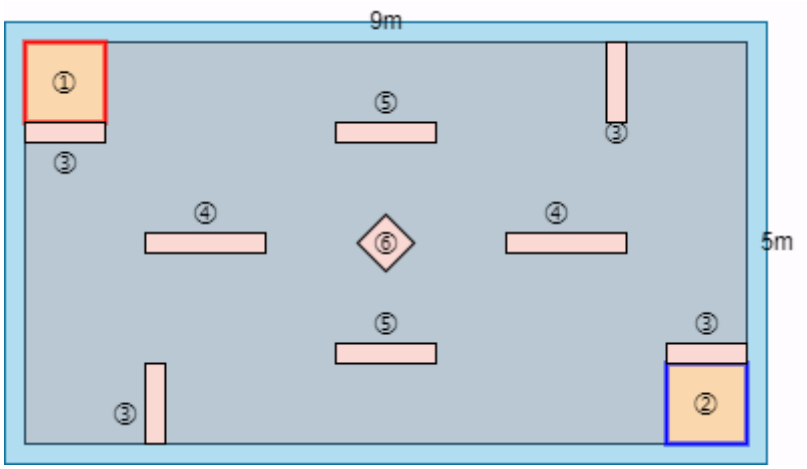

Figure 2. The map of environment

3. Research Plan & Expected outcome

The research process can be divided into several processes:

In the first stage, we will revisit related works including applications of DQN and the implementation of game theory in multi-agent decision making processes. Methods involving behavior tree methods will also covered since the behavior of agents can be more complex than deciding moving directions, the velocity should also be considered. To construct the competitiveness, we will also construct

In the next stage, we will construct an ideal environment where no inertia will be considered, thus the agents can stop or move as far as the decision of "stop" or "move in xm/s" is made. This stage is used to test the strategy of forming cost maps for each agent according to the threats of hostile agents. What's more, the agents are adopting single-agent strategies, which means there will not be any cooperation process.

In the third process, the agents will learn to cooperate. By adopting the cooperative strategies, the agents will learn to care about their allies and achieve overall victories, which means letting the team win instead of keeping themselves alive.

In the fourth process, to make the environment more similar to the reality, we will take Newton's laws of motion into consideration, which means the agents will be affected by inertia. The agents will have to use velocity lookahead algorithms (Zhang et al., 2011) in order not to miss the target point they have to reach.

As an outcome, we can develop a motion planning strategy based on DQN which can guide agents to cooperate and win human players. The strategy should also work in physical environments after modifying the trajectory of bullets.

References

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